

딥러닝 팀프로젝트

16조

1. Motivation & Goal

- 클라우드 의존 없는 on-device AI inference 수요가 커지지만, 모바일·노트북·엣지의 메모리·전력 예산은 한정적
 - ResNet18 (CIFAR-100 head) $\approx$ 11.2 M params, 44 MB FP32
- 해결 아이디어: Binary Neural Network (BNN)
 - Weight·Activation 을 ± 1 (1-bit) 로 양자화 $\rightarrow$ 이론 32 $\times$ 메모리 압축
 - MAC 연산을 XNOR + POPCNT 로 대체 $\rightarrow$ 정수·HW 가속 친화적
- 본 프로젝트 목표
 - CIFAR-100 W1A1 BNN 학습
 - Teacher 를 ablation study 로 강화 후 KD 로 student 정확도 향상
 - POPCNT 기반 C 추론 커널로 CPU (Intel Core i5-1135G7) 에서 speedup 실측

2. Method Overview

Why ReActNet-lite?

- ResNet18 backbone — ImageNet pretrained 활용
- ReActNet 의 학습 가능한 sign threshold (RSign) + PReLU 활성화 (RReLU)
- Bi-Real 의 double-skip 으로 1-bit 정보 손실 완화

모델 구성

- Teacher: FP32 ResNet18 (CIFAR-adapted, ImageNet pretrained → FT)
- Student: A1W1ReActNet (W1A1, FP32 stem + classifier, binary body)

4. Teacher: Ablation Study (7-step ablation)

- ResNet18 baseline (150 ep from-scratch): 75.74 %
- Step 1 architecture (7×7 stride 2 → 3×3 stride 1, maxpool 제거): 81.82 %
- Step 2 optimizer (SGD → AdamW, wd 1e-4): 82.38 %
- Step 3 batch / LR sweep: best 82.38 %
- Step 4 scheduler (CosineAnneal + warmup): 82.38 %
- Step 5 activation: ReLU 82.38 % (GELU 82.46 % — pretrained 호환성 우선)
- Step 6 mixup decay: 82.54 %
- Step 7 RandAugment mag=9: 82.58 % (mag=10 83.12 % — over-reg 보수 선택)
- Final Phase A: ImageNet pretrained + 80 epoch FT → 82.44 %

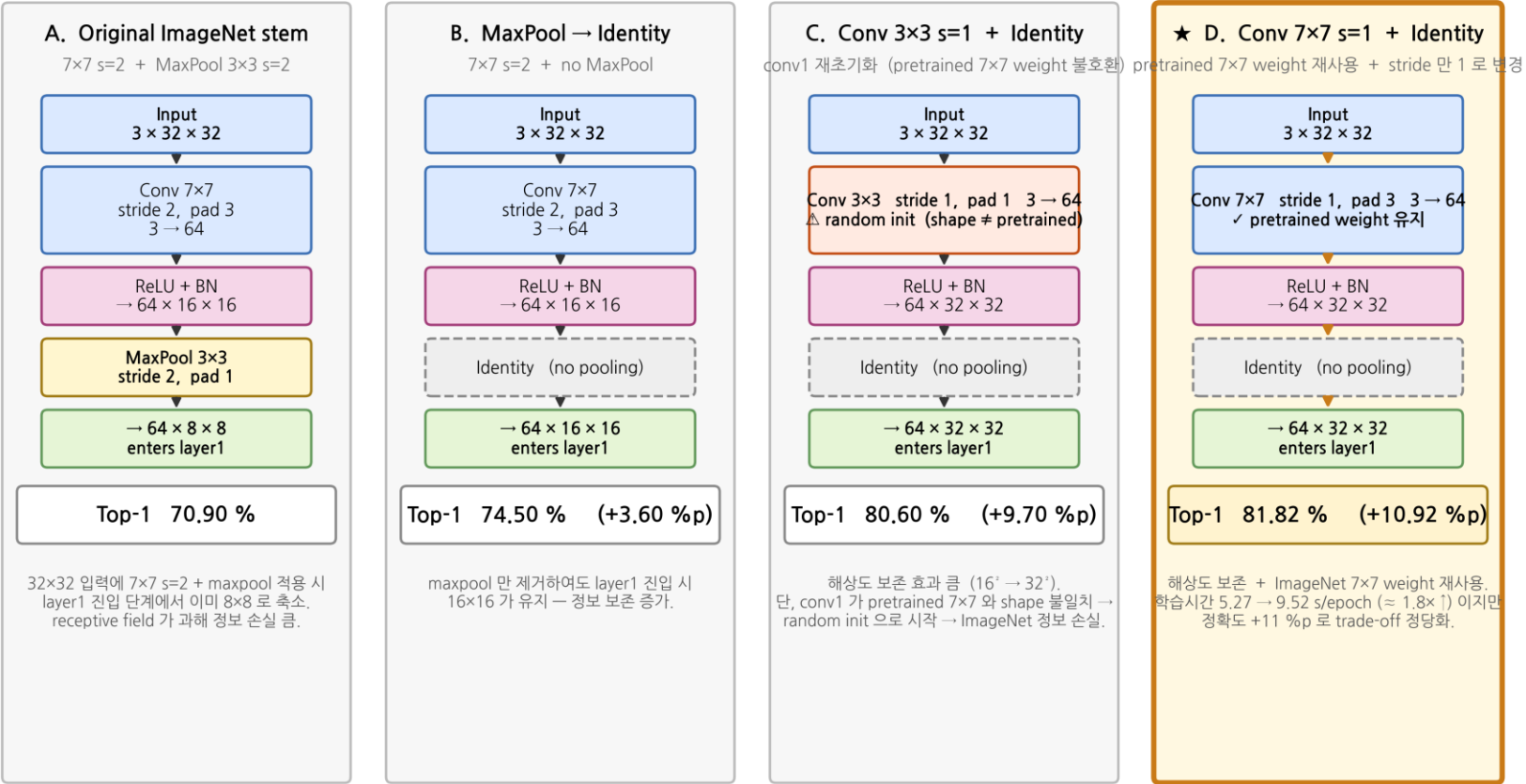
항목	값
Backbone	ResNet18 (torchvision, ImageNet pretrained)
Stem	Conv1 7 × 7 stride 1, maxpool → Identity
Optimizer	AdamW (wd = 1e-4)
Scheduler	LinearWarmup (8 ep, 0.1→1.0) + CosineAnnealingLR (72 ep, $\eta_{\min}$ = 1e-6)
Activation	(pretrained backbone 그대로) ReLU
LR / Batch / Epoch	1e-3 / 1024 / 80
Augmentation	RandAugment num_ops = 2, magnitude = 9, RandomErasing p = 0.2
Regularization	Label smoothing 0.05, Mixup α = 1.0 (step decay → $\alpha_{\min}$ = 0.0625), CutMix
Final top-1	82.44 %

4. Teacher: Ablation Study (7-step ablation)

- Step1 - Architecture

Step 1 — Stem adaptation (ImageNet ResNet18 → CIFAR-100 32×32)

larger spatial map entering layer1 ⇒ more usable features for 32×32 inputs (FP32 conv1 throughout — teacher phase)



변형	Top-1
Original (7 × 7 / s2, maxpool)	70.90 %
Maxpool → Identity	74.50 %
Conv1 3 × 3 / s1, maxpool → Identity	80.60 %
Conv1 7 × 7 / s1, maxpool → Identity	81.82 %

Takeaway maxpool 제거 (+3.6 %p) < stride 2 → 1 (+6.1 %p) → spatial-resolution 보존이 dominant factor. pretrained 7×7 weight 재사용 시 +1.22 %p 추가 이득.

4. Teacher: Ablation Study (7-step ablation)

- Step2 - Optimizer

Optimizer	Top-1
AdamW wd = 5e-2	81.50 %
AdamW wd = 1e-2	81.82 %
AdamW wd = 1e-3	82.20 %
AdamW wd = 3e-4	82.10 %
AdamW wd = 1e-4	82.38 %
Adam wd = 1e-4	81.44 %
SGD m = 0.9 wd = 5e-4 lr = 1e-1	81.20 %
SGD m = 0.9 wd = 5e-4 lr = 5e-2	81.86 %

4. Teacher: Ablation Study (7-step ablation)

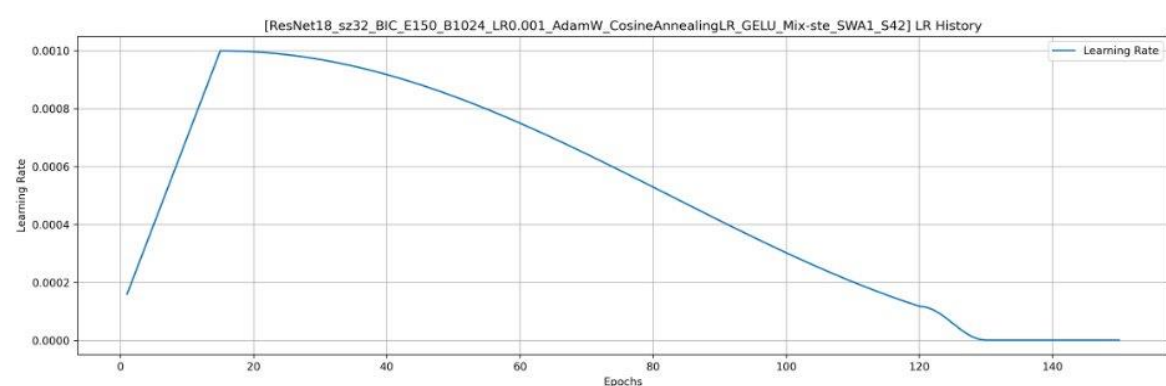
- Step3 – Batch size / Learning Rate

Batch	LR	Top-1
512	5e-4	82.42 %
512	7.07e-4	82.34 %
1024	1e-3	82.38 %
1024	2e-3	80.74 %
2048	1.41e-3	80.90 %

4. Teacher: Ablation Study (7-step ablation)

- Step4 - Scheduler

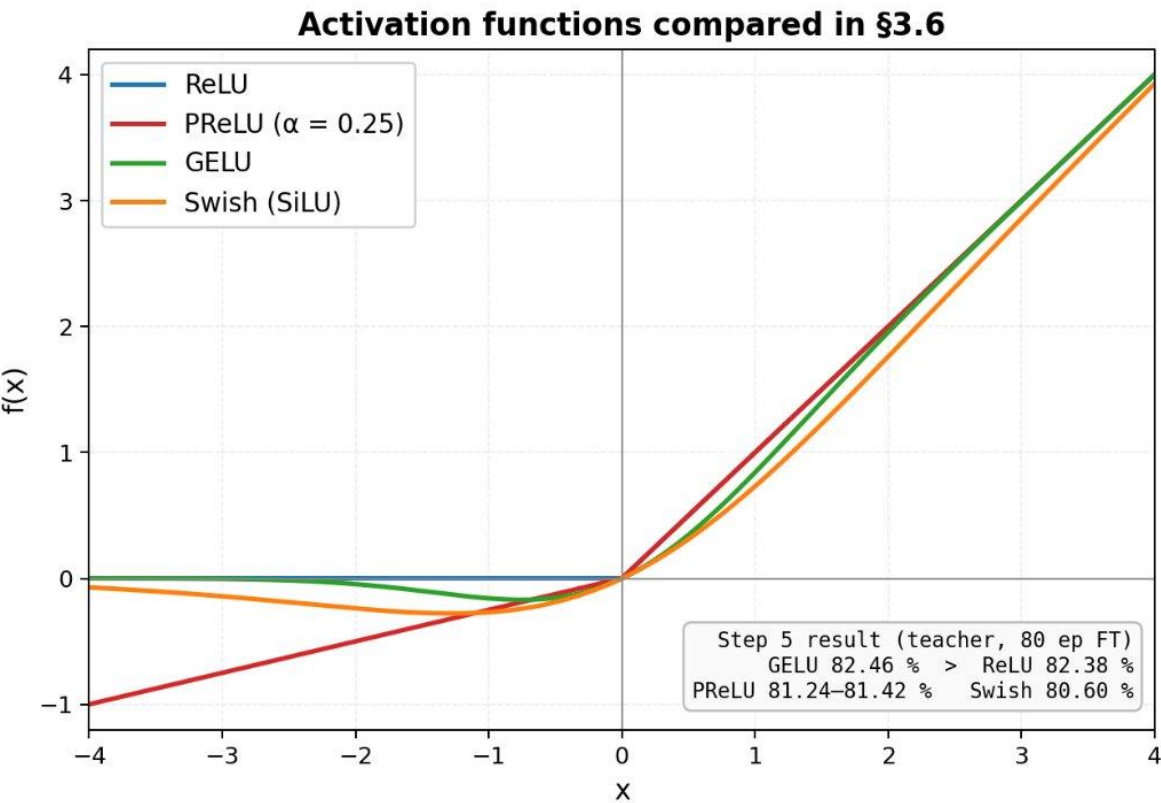
Scheduler	Top-1
CosineAnnealingLR	82.38 %
CosineAnnealingWarmRestarts ($T_0 = 20, T_{\text{mult}} = 2$)	81.22 %
MultiStepLR (60/110/140, $\gamma = 0.1$)	81.66 %
OneCycleLR	81.50 %



4. Teacher: Ablation Study (7-step ablation)

- Step5 - Activation

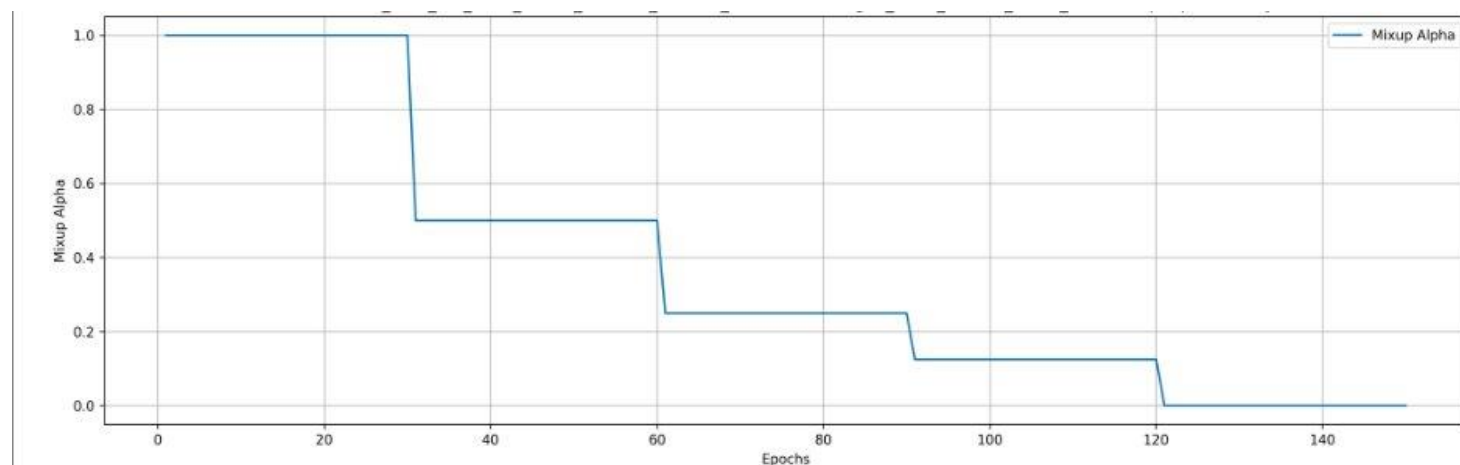
Activation	Top-1
ReLU (baseline)	82.38 %
PReLU per-layer	81.42 %
PReLU per-channel	81.24 %
GELU	82.46 %
Swish	80.60 %



4. Teacher: Ablation Study (7-step ablation)

- Step6 - Mixup Decay

Decay schedule	Top-1
Cosine	82.46 %
Linear	81.86 %
Constant	82.32 %
Step	82.54 %
Random	82.02 %
None	81.56 %



4. Teacher: Ablation Study (7-step ablation)

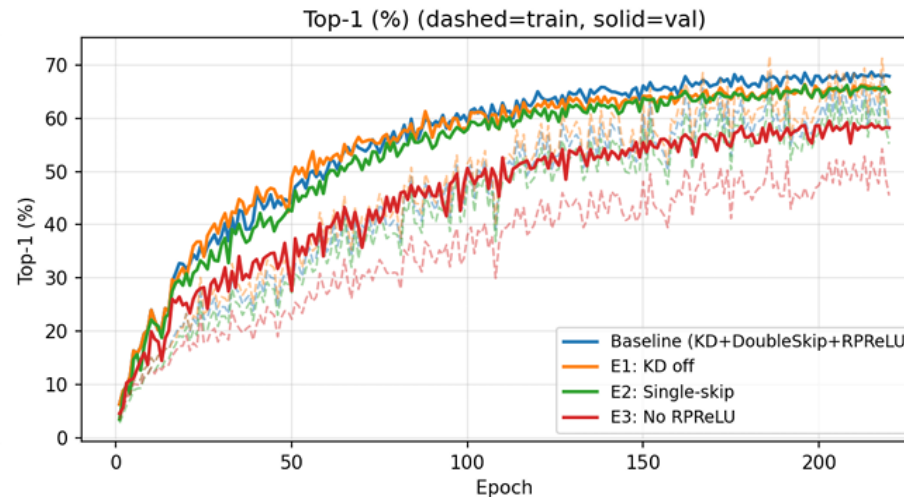
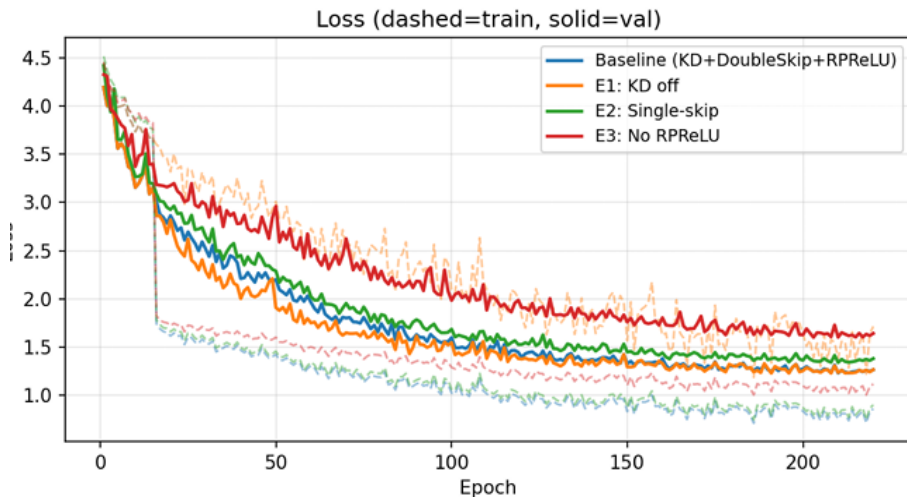
- Step7 - RandAugment

Augment	Config	Top-1
TrivialAugment	—	82.54 %
RandAugment	num_ops = 2, mag = 7	82.48 %
RandAugment	num_ops = 2, mag = 9	82.58 %
RandAugment	num_ops = 2, mag = 10	83.12 %
RandAugment	num_ops = 2, mag = 11	83.04 %
RandAugment	num_ops = 2, mag = 12	82.78 %
RandAugment	num_ops = 3, mag = 5	82.08 %
RandAugment	num_ops = 3, mag = 9	82.52 %

3. Student Architecture: A1W1ResNet18

- KD / Double-skip / RPreLU

Variant	Setting	Best Val Top-1	Final (ep 220)	Δ from baseline
Baseline	KD on + double-skip + RPreLU	68.74 % (ep 215)	67.94 %	—
E1: KD off	student_kd_weight = 0.0	66.44 % (ep 196)	64.96 %	-2.30 %p
E2: Single-skip	use_double_skip = False	66.28 % (ep 196)	64.94 %	-2.46 %p
E3: No RPreLU	block 의 RPreLU → nn.Identity() (제거)	59.52 % (ep 203)	58.26 %	-9.22 %p

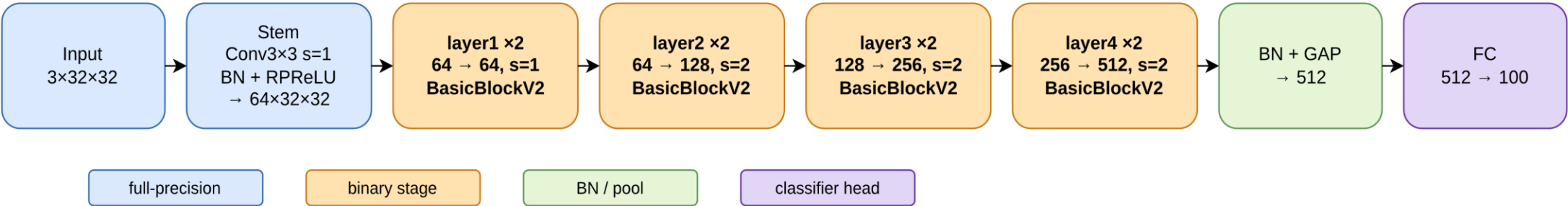


3. Student Architecture: A1W1ResNet18

ReActNet 의 핵심 트릭 (RSign + RReLU + double-skip + detached α) 차용 + 두 가지 단순화:

- backbone: ReActNet-A → 표준 ResNet18
- 학습: 2-stage W32A1→W1A1 KD → 단일 stage W1A1 KD

Binary Residual Block (double-skip + RReLU)

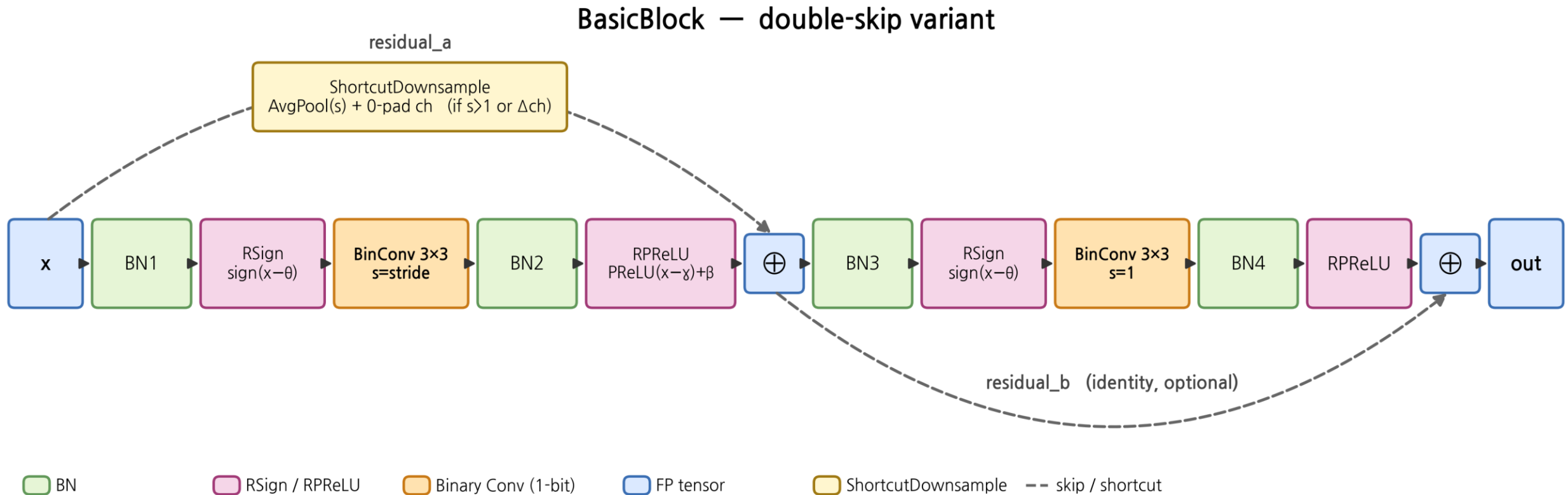


Memory budget (11.22 M params)

부분	params	bits/param	memory
Binary conv body (16 layers)	11.1 M	1 bit	1.39 MB
FP32 stem + classifier (1%)	0.12 M	32 bit	0.48 MB
Total student	11.22 M	—	≈ 1.4 MB
Teacher (FP32)	11.22 M	32 bit	44.0 MB

3. Student Architecture: A1W1ResNet18

- Model: ReactNet (Basic Block Architecture)



3. Student Parameters

분류	항목	Phase A	Phase B
Optimizer	AdamW	wd = 1e-4	wd = 0
LR	base / min	1e-3 / 1e-6	5e-4 / 1e-6
Scheduler	Warmup + Cosine	warmup 8 / cos 72	warmup 15 / cos 205
Epoch / Batch	—	80 / 1024	220 / 1024
KD	Logit (T, w)	—	T=4, w=0.7 (epoch ≥ 15)
EMA	shadow	decay = 0.999	disabled (binary 특성)
Mixup	α 초기 / min / decay	1.0 / 0.0625 / step	0.4 / 0 / step
Reg	label smoothing	0.05	0
Reg	grad clip	—	5.0

4. Training Pipeline

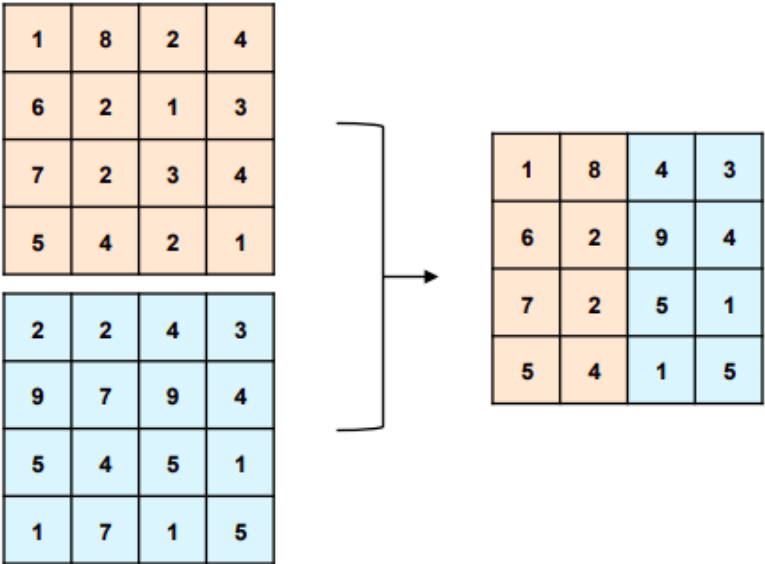
Level	기법	Param	역할
Image	RandomCrop	size 32, pad 4 (reflect)	위치 다양성
Image	RandomHorizontalFlip	p = 0.5	좌우 대칭
Image	RandAugment	num_ops = 2, mag = 9	색·기하 augment
Image	RandomErasing	p = 0.2, scale (0.02, 1/3)	occlusion robustness
Sample	Mixup	$\alpha = 1.0$ (teacher) / 0.4 (student), step decay	interpolation regularization
Sample	CutMix	$\alpha = 1.0$, prob 0.5	spatial replacement
Norm	Normalize	ImageNet stats	pretrained 호환성

4. Training Pipeline

Mixup



Cutmix



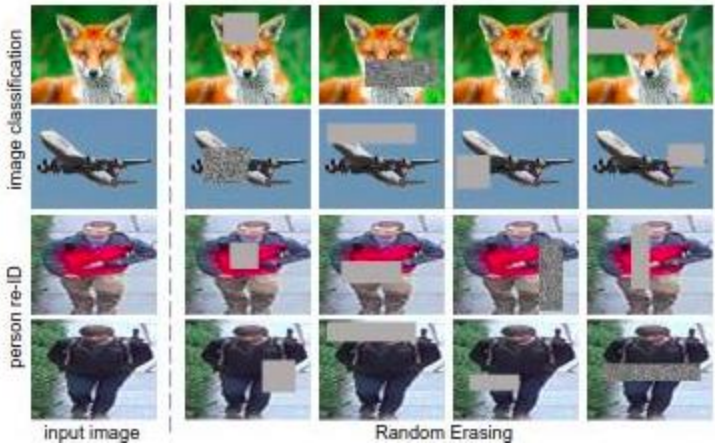
4. Training Pipeline

RandomErasing

1	8	2	4	2
6	2	1	3	5
7	2	3	4	7
5	4	2	1	4
9	7	1	5	2

→
RandomErasing

1	8	2	4	2
6	2	1	3	5
7	2	0	0	7
5	4	0	0	4
9	7	1	5	2



RandomHorizontalFlip

1	8	2	4	2
6	2	1	3	5
7	7	3	0	7
5	4	2	1	4
9	7	1	5	2

→
RandomHorizontalFlip
50%

2	4	2	8	1
5	3	1	2	6
7	0	3	7	7
4	1	2	4	5
2	5	1	7	9

4. Training Pipeline

Single Kaggle session (T4, 12h budget)

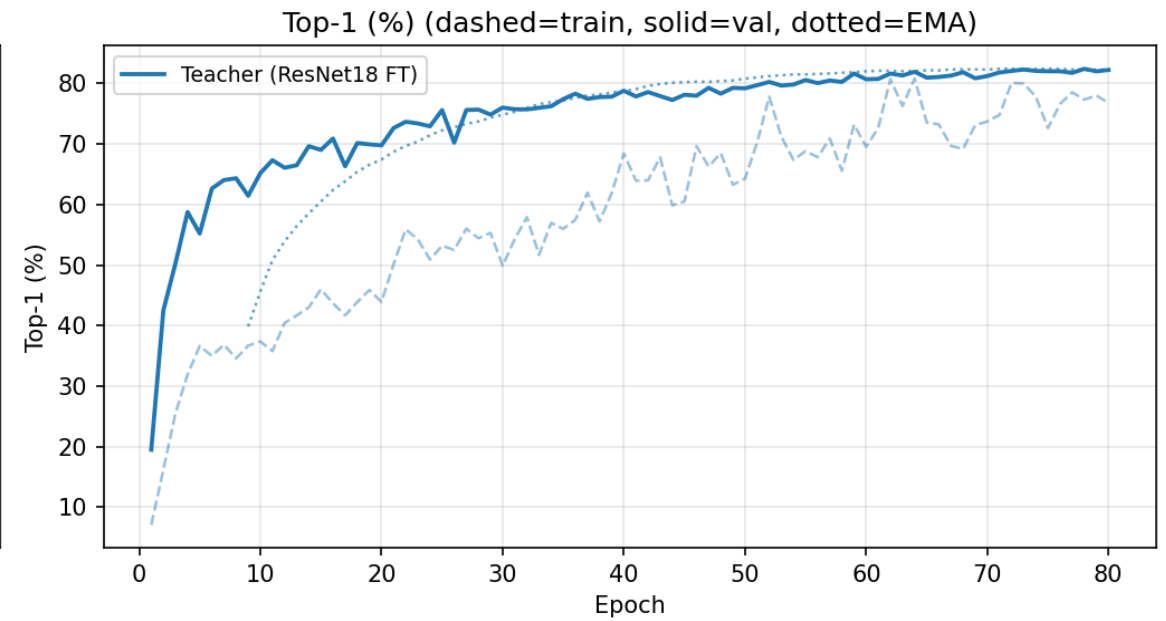
Phase	Model	Epoch	Time
A	Teacher FT (ImageNet → CIFAR)	80	≈ 15 min
B	Student (A1W1 + Logit KD T=4, w=0.7)	220	≈ 80 min
Total			≈ 1h 35 min

Phase B 핵심 hyperparam

- Optimizer: AdamW (wd = 0), LR $5e-4 \rightarrow 1e-6$ cosine
- Warmup 15 ep, KD 활성화 시점 = warmup 종료 (epoch 15)
- Latent weight clamp [-1, +1] (매 step) — sign STE dynamic range 보호
- Augmentation: RandAugment / Mixup $\alpha=0.4$ step decay / CutMix / RandomErasing

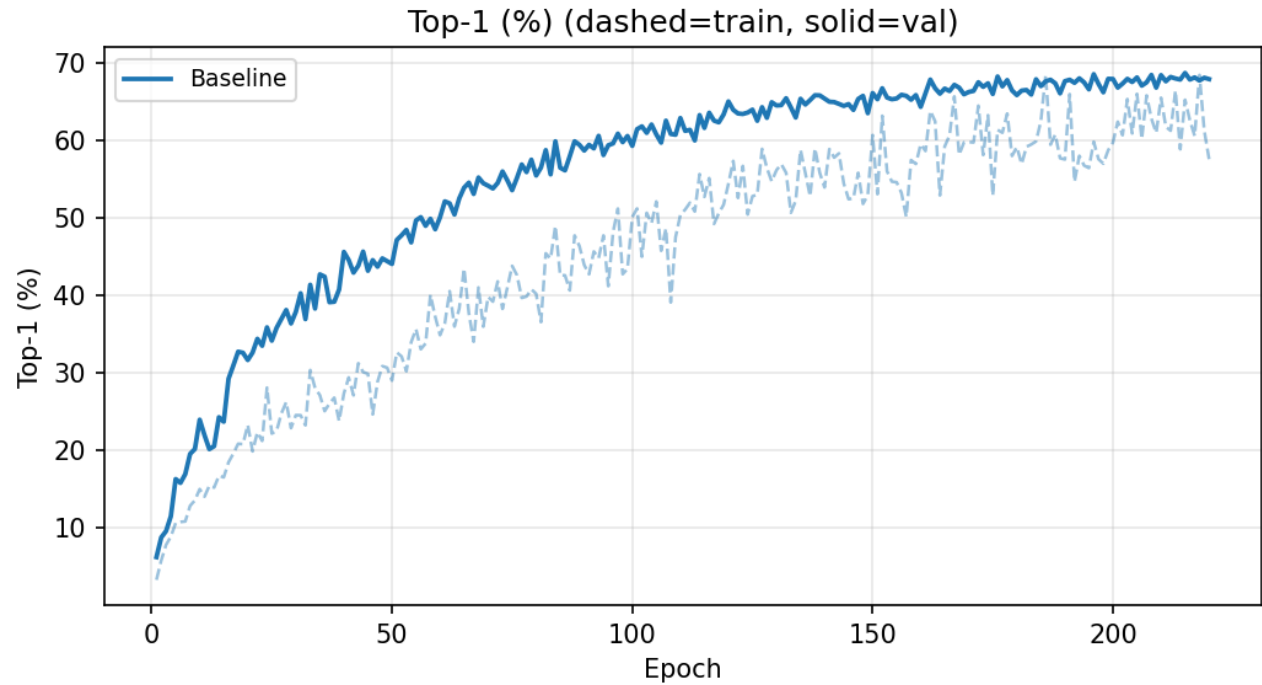
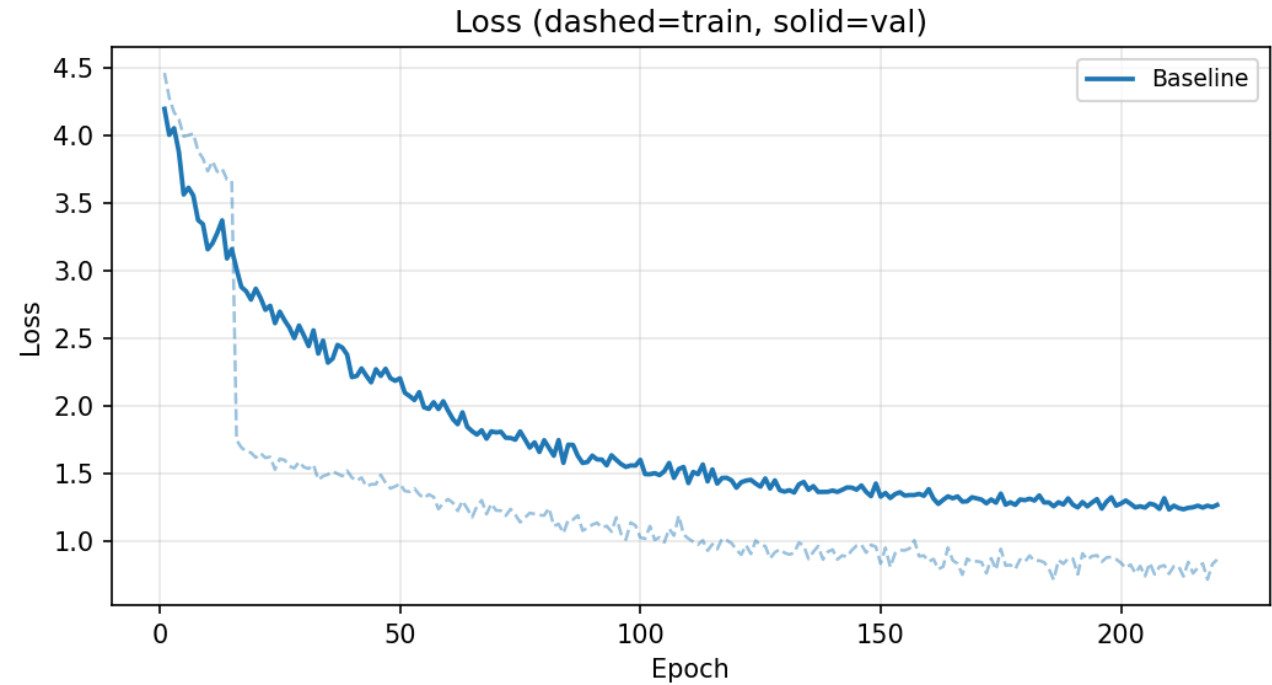
5. Training - Teacher

Phase A — Teacher Training Curves (80 epochs)



5. Training - Student

Baseline Student — Training Curves (220 epochs)



6. Quantitative Results

CIFAR-100 Top-1

Model	Params	Memory	Top-1
Teacher (FP32 ResNet18)	11.22 M	44.0 MB	82.44 %
Student (A1W1 ReActNet-lite)	11.22 M	1.4 MB	69.06 %

Literature comparison (W1A1)

Method	Dataset	Top-1
XNOR-Net (ECCV'16)	ImageNet	51.2 %
Bi-Real (ECCV'18)	ImageNet	56.4 %
ReActNet (ECCV'20)	ImageNet	69.4 %
ReCU* (ICCV'21)	ImageNet	66.4 %
RBNN	CIFAR-100	~ 67.1 %
Ours	CIFAR-100	69.06 %

7. Design Choices & Ablation

- FP32 stem + classifier (1%): 200 KB 만 차지하면서 +5–10 %p 정확도 보존 (BNN 표준)
- AMP fp16 binary STE: ± 1 sign 은 fp16 동적 범위 내 안정, Tensor Core 활용 $\rightarrow$ 학습 1.5–2 $\times$
- Ablation 검증 — 컴포넌트 기여도 (Backup F 상세):
 - RReLU 가 단일 최대 기여 — 제거 시 -9.22 %p
 - Double-skip 제거 시 -2.46 %p, KD off 시 -2.30 %p (비슷한 크기)

8. C Inference: Design + Speedup

Wall-clock latency (CPU: Intel Core i5-1135G7, 4C / 8T)

Core	Student (W1A1)	Teacher (FP32)	Ratio
1	24.65 ms	522.93 ms	21.21 ×
4	11.87 ms	150.49 ms	12.68 ×

왜 thread 수에 따라 ratio 가 변하는가

- Teacher (FP32): 큰 GEMM 루프 → multi-thread 로 $\sim 3 \times$ 가속
- Student (binary): 작은 op 16 개 (binary conv) → thread spawn overhead 로 $\sim 1.5 \times$ 만 가속
→ 4-thread 시 teacher 가 더 큰 이득 → ratio 21 → $12 \times$ 로 감소

8. C Inference: Design + Speedup

FP32



ReActNet



9. Limitations & Honest Discussion

- T4 GPU INT1 native 미지원 (Turing 은 INT4 까지) → GPU 가속 불가, CPU 추론 demo 로 우회
- Naive C 비교 한계:
 - 메인 C 는 VPOPCNTDQ / SIMD-GEMM 미사용 → $25\times / 12\times$ = 알고리즘 차이
- Ablation 범위 한계 :
 - 단일 컴포넌트 ablation 만 수행 — RPreLU 가 최대 기여 (-9.22 %p)
 - cross-component interaction, KD weight sweep 미수행

10. Conclusion + Future Work

핵심 성과

- CIFAR-100 W1A1 BNN 69.06 % top-1
- 메모리 32× 압축 (44 MB → 1.4 MB) + 체계적 teacher empirical study (75.7 % → 83.1 %)
- POPCNT-based C 추론 demo — PyTorch 와 bit-exact, single-thread 21× speedup 실측

Thank You

Team 16 · KAU Deep Learning Term Project